Middle-Rung Law: Predictions of a Three-Rung Frequency Ladder and Formula Compendium

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Normalization relation $4\pi^2r^3c^2/(T^2Ghv)=1$ written by Zhang Xiangqian (2026-03-03). All numerical values use CODATA 2018 constants.

Part I — Preregistration and Falsifiable Predictions

1. Hypothesis

The Planck frequency is the geometric mean of the cosmic expansion rate H and an ether frequency ν_e: at every epoch t, $\nu_e(t) \cdot H(t) = \nu_P^2$, equivalently $\hbar G = c^5/(4\pi^2\,\nu_e\,H)$. ν_e equals the de Broglie angular frequency of the Hubble sphere treated as one object (Kepler's third law with $r = c/H$, $T = 2\pi/H$, combined with $Mc^2 = h\nu$; normalization relation written by Zhang Xiangqian, 2026). Prior to cosmic age 3.912 Gyr (3.641–3.950 Gyr across H_0 = 61.8–63.5 km s^{-1} Mpc^{-1}), the early universe undergoes hot radiation/matter evolution and intense hierarchical merging of smaller structural units; at **9.900 Gyr ago** (range 9.893–9.933 Gyr ago, redshift $z_s = 1.686$, range 1.669–1.817; acoustic-angle benchmark $z_s = 1.721$ gives age 3.837 Gyr and scale ratio $1+z_s = 2.721 \approx e$), large-scale galaxy structures stabilize and cosmic expansion locks into uniform coasting $H(z) = H_0(1+z)$, after which $\nu_e(t)$ rises linearly with cosmic age while H(t) declines inversely with fixed product $\nu_P^2 = 8.715\times10^{84}$ Hz².

2. Predictions

ID	Quantity	Predicted value	Deciding data	Falsified if
P1	Hubble constant H_0	61.8–63.5 km s^{-1} Mpc^{-1}	Gravitational-wave standard sirens (free of astrophysical calibration)	Measured value outside range at 3σ (e.g. 73)
P2	Age of universe	13.5–13.9 Gyr (central 13.82 Gyr)	Oldest stellar ages + independent cosmological determinations	Reliably < 13.4 Gyr or > 14.0 Gyr
P3	Expansion history & coasting transition	Transition to uniform coasting $H(z) = H_0(1+z)$ at 9.900 Gyr ago (9.893–9.933 Gyr ago, age 3.912 Gyr, $z_s \approx 1.69$); at z = 2.33, H is ~3% below ΛCDM	DESI final, Euclid, and Roman BAO measurements	z = 1.5–2.5 consistent with ΛCDM at < 1% precision with no transition
P4	Dark matter	Early compact extreme objects, 10^{14}–10^{20} kg	Microlensing, femtolensing, sub-solar gravitational-wave searches	Mass window completely excluded
P5	MOND-like acceleration scale	$a_0 = (0.96–0.98)\times10^{-10}$ m s^{-2} today, growing with H(z): ×2 at z = 1, ×3 at z = 2	High-redshift disk galaxy rotation curves	No redshift evolution
P6	The Planck gate (non-classical motional limit)	No mechanical degree of freedom with effective mass $m_{eff} \geq m_P$ = 21.8 μg (total frequency ≥ $\nu_P = 2.952\times10^{42}$ Hz) can exist as a wave at the human rung or be prepared in a non-classical motional state	High-overtone bulk acoustic resonators + superconducting qubits, levitated particles, massive optomechanics	A peer-reviewed experiment prepares ≥3σ Wigner negativity or a cat-state superposition (separation > 2× zero-point amplitude) for $m_{eff} \geq$ 21.8 μg
P6.1	The Planck blind band & observational collective effect	Within 21.8 ± 0.3 μg (21.5–22.1 μg baseline, wider under strong collective coupling), field-pressure readout between the target and the observational collective decouples: both classical and non-classical observations vanish (<1% of out-of-band baseline, resuming outside), with window centre and width shifting reproducibly across laboratories, nearby heavy equipment, and observer body-mass frequencies	Tunable bulk-acoustic / optomechanical resonators spanning 16–25 μg, microgram levitated-particle scattering scans, and differential runs across laboratories, heavy equipment, and manned vs. unmanned operation	A peer-reviewed scan across 21.0–22.6 μg observes smooth, continuous classical optical/mechanical scattering and readout throughout the entire interval (<10% attenuation)

Derived quantities: $N = \nu_P/H = (1.44–1.47)\times10^{60}$; $\nu_e = (4.2–4.4)\times10^{102}$ Hz; mid-rung frequency $\sqrt{(H\,\nu_P)}$ = 2.43–2.46 THz. Ether-flow (gravitational potential) variations add in quadrature; under current axioms, a dual-optical-clock comparison next to any device without mass change should read zero.

2.1 Operational Criteria and the Observational Collective Mechanism for P6 and P6.1

Physical mechanism (the Observational Collective): The Planck mass $m_P = \sqrt{(\hbar c/G)} = 21.8$ μg has total frequency $m_P c^2/h$ equal to the human-boundary frequency $\nu_P = 2.952\times10^{42}$ Hz. By Axioms 1, 3, and 4, an observation is never an isolated object acting alone: the target matter (near 21.8 μg), the probing light, the vacuum, the measuring apparatus, the observer's eye and body (whose body mass M_{obs} carries its own ether frequency $F = M_{obs}c^2/h$), and the surrounding laboratory building and nearby heavy equipment are all distinct ether structures with their own frequencies and overlapping ether spaces. Human observation is the field-pressure coupling ($v^2 = 2\Phi$) across this entire **observational collective**. When the effective mass of a target mode enters the neighbourhood of ν_P (21.8 μg), the collective frequency of the coupled system gates human-layer readout at ν_P: field-pressure transduction decouples, causing **both classical and non-classical observations to vanish simultaneously** within the Planck blind band (P6.1); above the upper edge of the blind band (> 22.1 μg), classical field-pressure readout resumes, but because the total frequency exceeds ν_P, the structure can no longer exist as a wave at the human rung and no non-classical motional state can be formed (P6).

Criterion	Fixed operational definition
Effective mass m_{eff}	The effective (motional) mass of the mechanical/vibrational mode as reported by the experiment; for centre-of-mass motion or levitated particles, the total mass of the target.
P6 non-classical state criteria (either suffices)	(i) Wigner-function negativity of the motional state at ≥ 3σ significance; (ii) a superposition of two coherent states (cat state) separated in phase space by more than twice the zero-point amplitude, evidenced by interference fringes or Wigner negativity.
P6.1 dual-channel disappearance criteria (both required)	(i) Inside the target's collective blind window (baseline 21.8 ± 0.3 μg), the effective scattering cross-section or readout response across all photon-, phonon-, or field-pressure-mediated classical (optical scattering, thermal emission, centre-of-mass readout) and non-classical channels drops to **< 1% of its out-of-band baseline (or below instrument noise floor)**; (ii) tuning m_{eff} or measuring control samples outside the window (< 21.5 μg or > 22.1 μg) immediately restores normal classical scattering and mechanical readout (ruling out sample loss or damage).
Why it is an interval & collective environmental shift	Because the decoupling threshold is governed by the superposed collective frequency field of the target, probing light, vacuum, detector, observer's eye and body ($F_{body} = M_{body}c^2/h$), laboratory building, and surrounding heavy apparatus, the blind region is a finite band rather than a mathematical point at 21.764 μg. Repeating the measurement on the same target across **different laboratories, toggling or relocating nearby large-scale equipment (high-field magnets, heavy machinery, high-power RF sources), or varying the presence and body masses (frequencies) of near-field human observers** produces a **reproducible shift in both the centre mass and width of the blind window**.
Non-qualifying cases	For P6: near-ground-state cooling, squeezing, or Gaussian entanglement with everywhere-positive Wigner functions (e.g. near-ground-state cooling of the 10 kg LIGO mirrors is not a counterexample). For P6.1: signal loss from sample evaporation, mechanical detachment, optical misalignment, cavity detuning, or ordinary detector malfunction.
Expected trend (not a falsification condition)	Approaching the collective blind band from either side produces a steeper edge drop in both classical signal-to-noise ratio and non-classical cat-state coherence time than environmental decoherence models predict; the dip window widens when heavy surrounding equipment is active or multiple observers are in close proximity compared to unmanned remote telemetry.

3. Disclosure of Data Already Examined

Experimental and observational datasets examined prior to or upon fixing these predictions:

- **P5:** Ciocan et al. 2026 (*Astronomy & Astrophysics* 709, L16) report a_0 rising with redshift, reaching 2.38×10^{-10} m s^{-2} at $z \approx 1$. The direction agrees with P5; the magnitude is ~19% above our prediction. Earlier analyses (Milgrom 2017; Limbach et al. 2008) disfavour strong evolution.
- **P3:** DESI DR2 BAO data (13 points). Our model gives best $\chi^2 = 27.7$ (fixed acoustic-angle calibration; 21.64 vs. 10.54 for ΛCDM when r_d is free under identical degrees of freedom), which is unfavourable to our model, with the largest residual at $z = 0.934$.
- **P1:** Current gravitational-wave-only standard-siren bounds (±15–20 km s^{-1} Mpc^{-1}) do not yet discriminate.
- **P6 & P6.1:** The heaviest mechanical mode prepared in a non-classical state to date is Bild et al. 2023 (*Science* 380, 274), demonstrating cat states at $m_{eff} = 16.2$ μg (74% of the 21.8 μg Planck gate); macroscopic particle scattering experiments have not yet performed high-resolution continuous mass-stepped coupling scans across the 21.5–22.1 μg window. Standard physics predicts neither a non-classical limit nor a dual-channel blind band at 21.8 μg, making both P6 and P6.1 directly falsifiable within months to years.

Every future comparison, favourable or not, will be recorded without altering the predictions above.